

A Physics-Informed Grey-Box Model of an Oscillating Heat Pipe: Identification, Extrapolation, and Working-Fluid Coupling Ratios

Jaminur Rahman Khalid

Abstract

Choosing a working fluid for oscillating heat pipes (OHPs) usually requires a separate experiment per candidate, and the underlying two-phase flow resists first-principles modelling. We identify a physics-informed grey-box model of a helically coiled OHP from transient temperature measurements of one apparatus operated with ethanol, methanol and deionised water, each in three heating runs (vessel plateaus ~ 58 , ~ 70 , ~ 93 °C; nine run–fluid series). A lumped two-node energy balance, driven by the measured vessel temperature, is fitted by a physics-informed neural network (PINN), which learns the physical parameters jointly with the temperature curves through the ODE residual, and equally by plain ODE shooting. Under leave-one-run-out validation the identified model predicts the evaporator temperature of an unseen run at 0.24 K to 0.46 K when interpolating and 0.71 K to 0.85 K when extrapolating (up or down) in heating level. Reference baselines given the same measured initial state locate what this tests: a trivial constant-gap predictor stays within 3 K; nonlinear black boxes (multilayer perceptron, Gaussian process, degree-2 SINDy) reach 0.6 K to 6 K on the interpolation fold but fail at 24 K to 87 K on extrapolation, with a recurrent GRU best among them at 5 K; a one-node ablation of the same ODE matches the two-node fits on the evaporator, so the second node is justified by the condenser and adiabatic channels, not by evaporator accuracy; and an unconstrained per-fluid linear state-space with no physical parameterisation matches or beats the physical model in every fold – the linear structure, fitted by output-error shooting, is what carries extrapolation, and the physical parameterisation is retained for parsimony and interpretability. Plain ODE shooting is slightly better than the PINN in every fold: for this fully observed 9-parameter linear system the structure, not the optimiser, carries the accuracy. A dimensionless coupling ratio $R^* = a_f/\kappa_f$, identifiable also from plateau ratios, ranks the fluids in every fit (EOHP ≈ 12.6 – $13.5 >$ MOHP ≈ 7.1 – $7.3 >$ WOHP ≈ 5.8 – 6.0); its $\sim 48\%$ drift for ethanol across heating levels is consistent with the ethanol dry-out reported in the source study, a regime the constant-parameter model cannot represent; a per-run temperature-dependent $\kappa(T_e)$ is identifiable but does not transfer across heating levels, so the deployed model keeps κ constant. The fitted right-hand side exports as ONNX; scenario sweeps within the validated envelope illustrate model-implied trade-offs between setpoint tracking and ripple rejection along R^* , and are explicitly not fluid recommendations. Limitations: one geometry, one shared driver per run, slow ramps, 5 s sampling, equal vessel–evaporator coupling assumed, and no heat-flux or dry-out limits.

1 Introduction

Oscillating heat pipes (OHPs) achieve high effective conductivity through self-excited two-phase oscillation, but the same mechanism makes their behaviour difficult to model from first principles: the flow regime, the distribution of liquid slugs and vapour plugs, and the evaporation–condensation dynamics are only partially observable through wall temperatures. Working-fluid comparison is therefore usually experimental: one apparatus per fluid, per fill ratio, per geometry. Data-driven surrogates trained on such measurements interpolate well but are known to fail when asked to

extrapolate to unseen operating conditions – precisely the question a designer asks (“what if we run hotter?”).

Grey-box identification offers a middle road: impose the physically mandatory structure (energy conservation on a small number of lumped nodes) and learn the few effective parameters from data. Physics-informed neural networks (PINNs) provide a convenient machinery for this, embedding the ODE residual in the loss through automatic differentiation and optimising physical parameters jointly with a trial solution. What PINNs add over classical shooting-based parameter fitting for such small, well-posed ODEs is, however, an open and often glossed-over question.

This paper contributes a deliberately honest case study on a public dataset of a helically coiled OHP (HCOHP) [1, 2] operated with three working fluids:

1. a two-node grey-box model, vessel-temperature-driven, identified by PINN and by ODE shooting, validated leave-one-run-out with extrapolation both up and down in heating level, against IC-fair baselines: trivial, linear, and nonlinear black boxes (MLP, GP, SINDy, autoregressive GRU);
2. evidence that a correctly structured linear model, not the PINN optimiser or the physical parameterisation, carries the accuracy: shooting, one-node, and even an unconstrained per-fluid linear state-space extrapolate at 0.3 K to 0.9 K while nonlinear black boxes degrade (GRU 5 K, others 24 K to 87 K; trivial gap baseline 3 K);
3. a dimensionless coupling ratio $R^* = a_f/\kappa_f$ that is invariant to the sampling interval, seed- and method-stable, ranks the fluids identically in every fit, and is recoverable from plateau ratios alone – with the ethanol drift consistent with the dry-out reported in the source data;
4. negative results reported as such: a temperature-dependent pipe conductance $\kappa(T_e)$ is identifiable within each run (an initial “not identifiable” verdict was an optimiser artefact, corrected) but is a heating-level effect that does not transfer across runs, so the deployed model keeps κ constant; per-fluid condenser loss rates do not pay for their parameters;
5. a consistency check on an untrained channel: adiabatic-section temperatures are reproduced as a static mixture of the two identified nodes to ± 0.5 K on held-out EOHP/WOHP runs (MOHP is harder);
6. an exported ONNX simulator used for model-implied scenario illustrations (tracking vs. ripple rejection along R^*) with seed-ensemble bands – explicitly not fluid recommendations, since the model carries no heat-flux or dry-out limits.

2 Related work

Oscillating heat pipes. OHPs (pulsating heat pipes) are reviewed comprehensively in [4]; their effective behaviour emerges from self-excited two-phase oscillation that resists first-principles modelling, which is why thermal-performance studies largely remain experimental and apparatus-specific. Our data source, the HCOHP study of Yeboah and Darkwa [3, 1, 2], is unusual in publishing raw thermocouple records for three working fluids under three heating levels, and in reporting fluid-dependent dry-out onset (ethanol first) – the phenomenon our constant-parameter model structurally cannot represent, and which any fluid-selection use of these models must confront [5]. The common engineering metric is a steady thermal resistance $R_{th} = (T_e - T_c)/Q$ [5, 3]; our R^* is deliberately dimensionless (a ratio of identified rates, recoverable from plateau ratios) because absolute fluxes are not identifiable from these records.

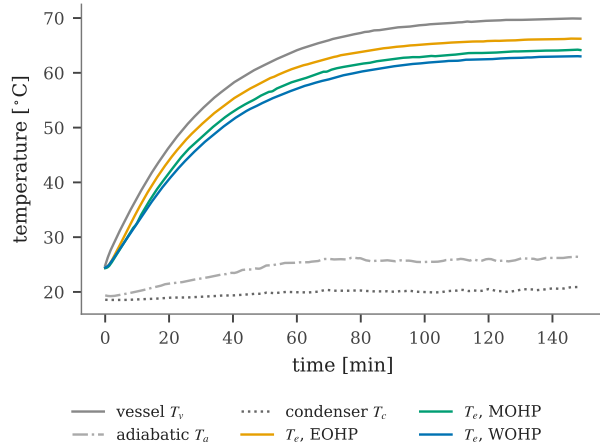


Figure 1: Raw thermocouple channel means for Run 2 (all three fluids); the vessel/adiabetic/condenser channels are shared across fluids within a run (common bath), so they are shown once.

Grey-box identification. Imposing physically motivated structure on a low-dimensional state-space and fitting its parameters from data is classical system identification [6]. The methodological points we lean on are old and robust: parameter sloppiness (stiff ratios, sloppy rates) is generic in multi-parameter physical models [7], which is why we report the ratio $R^* = a_f/\kappa_f$ rather than the rates; and structure selection must be validated out-of-sample, which is what the LORO protocol with two extrapolation directions provides.

PINNs for inverse problems. PINNs embed governing equations in the loss of a neural trial solution [8] and have become a standard tool for forward and inverse problems [9]. Known pathologies (gradient flow, optimisation sensitivity) are documented [10]; for small, fully observed lumped systems the classical alternative – least-squares shooting over the same equations – is trivially available, and controlled comparisons of the two on identical structure are rarer than method papers. Our result (shooting $\geq$ PINN in every fold, one-node $\approx$ two-node on the target channel) is a data point for that literature, in the spirit of baseline-honest evaluations.

Data-driven dynamics. SINDy [11] and GPs [12] represent the equation-discovery and nonparametric-regression routes; our GRU baseline the sequence-learning route. Their uniform failure on the extrapolation fold (while a trivial constant-gap predictor stays within 3 K) echoes the general point that interpolation skill does not transfer across operating regimes. The conformal prediction literature [13] offers distribution-free predictive bands; our seed ensembles quantify only parameter uncertainty, and we mark conformal augmentation as future work rather than claim calibrated bands.

3 Data

We use the public HCOHP dataset of Yeboah and Darkwa [1, 2]: a copper helical coil (2 mm ID tube, 8 cm coil diameter, 10 turns; evaporator/adiabetic/condenser sections 0.19/0.20/0.19 m) on a heated vessel ($L = 0.30$ m), instrumented with K-type thermocouples logged at 5 s. Three working fluids – ethanol (EOHP), methanol (MOHP), deionised water (WOHP) – were each tested in three heating runs reaching vessel plateaus of approximately 58, 70 and 93 °C (Fig. 1). Per (run, fluid) we form channel means: vessel T_v (outer wall, the driver), evaporator T_e (3 TCs), condenser T_c (3

TCs), adiabatic T_a (2 TCs). Models are fitted on Savitzky–Golay-smoothed, 25 s-subsampled series; all reported errors compare predictions interpolated onto the raw unsmoothed records.

4 Model and identification

4.1 Two-node grey-box structure

With the measured vessel temperature as exogenous driver, per fluid f :

$$\dot{T}_e = a_f (T_v - T_e) - \kappa_f (T_e - T_c), \quad \dot{T}_c = r \kappa_f (T_e - T_c) - d (T_c - T_{\text{cool}}), \quad (1)$$

where a_f is the vessel–evaporator coupling over evaporator heat capacity, κ_f the pipe conductance over the same capacity, $r = C_e/C_c$, d a condenser loss rate, and T_{cool} the sink temperature; r , d , T_{cool} are shared across fluids. The dimensionless quantity

$$R^* = \frac{R_{\text{pipe}}}{R_{\text{in}}} = \frac{a_f}{\kappa_f} \quad (2)$$

requires no knowledge of the evaporator heat capacity and is the natural per-fluid output of the identification. We assume the vessel–evaporator coupling geometry is common to the three pipes (a_f differences are fluid-side); within-run vessel inner/outer thermocouples are identical across fluid sheets, consistent with a shared bath measurement, and the plateau $T_v - T_e$ ordering matches the fitted a ordering.

4.2 PINN

A small trial network $U_\theta(t, \text{series}) \rightarrow (T_e, T_c)$ (3×64 tanh MLP) is trained on the smoothed thermocouple data plus the ODE residual of Eq. (1), evaluated on U by autodiff in time; the physical parameters ($\log a_f, \log \kappa_f, \log r, \log d, T_{\text{cool}}$) are optimised jointly (Adam, cosine decay, 6000 steps, float32). A state-dependent variant learns $\log \kappa_f += g_f(T_e, T_e - T_c)$ with a small MLP g . Ten seeds per fold quantify the fit’s parameter dispersion.

4.3 ODE shooting

The same Eq. (1) with constant κ_f is fitted by least-squares shooting (scipy `least_squares` over log-parameters, LSODA rollouts). This is the classical route; comparing it to the PINN isolates what the neural machinery contributes.

4.4 Black-box baselines (IC-fair)

All baselines receive the same information the physics fits use, including the measured initial state: (MLP, GP) map $(t, T_v, \dot{T}_v, \text{fluid one-hot}, T_e(0), T_c(0)) \rightarrow (T_e, T_c)$ pointwise; SINDy (STLSQ, degree-2 polynomial library in (T_e, T_c) with T_v and fluid one-hot as control, states std-scaled, time in hours) is integrated by clipped RK4 rollout from the measured initial state; the GRU baseline is an autoregressive dynamical model (GRU cell, hidden 64) consuming $(t, T_v, \dot{T}_v, \text{fluid}, T_e^{\text{prev}}, T_c^{\text{prev}})$, trained teacher-forced one-step-ahead (25 s steps) and rolled out from the measured initial state, driven by the measured $T_v(t)$. Three GRU seeds; GP with wide kernel bounds (10^{-3} – 10^4 , 5 restarts) is reported as GP.

Four further references probe what the comparison actually tests: **Trivial** $T_e = T_v - c_f$ (per-fluid mean gap); **Ridge-linear**, a linear state-space $(T_e, T_c) = W [T_e, T_c, T_v, \text{fluid}, 1]$ fitted by ridge

regression on finite-difference derivatives and rolled out by RK4 (the linear data-driven counterpart); a **one-node** ablation $\dot{T}_e = a_f(T_v - T_e) - d(T_e - T_{\text{cool}})$ fitted by the same shooting procedure (five parameters, no condenser node; predicts T_e only); and **Linear-SS**, an unconstrained per-fluid linear state-space $\dot{X} = AX + gT_v + c$ with all eight entries free per fluid (no cross-fluid or physical sharing), fitted by the same output-error shooting – a free linear form of the same order as the physical model.

4.5 Validation protocol

Leave-one-run-out (LORO): fit on six (run, fluid) series of two runs, predict all three fluids of the held-out run by rolling the identified ODE (or GRU) from its measured initial state, driven by its measured $T_v(t)$. Because the three fluids within a run share the same vessel drive, each fold has effectively one independent driver with three repeated measures. Only holding out Run 2 is interpolation; Run 1 is downward extrapolation (training on ~ 70 and ~ 93 °C, testing at ~ 58 °C) and Run 3 upward extrapolation (training ≤ 70 °C, testing ~ 93 °C). Structure-selection choices made during the analysis (constant κ , shared d , driver variant, T_c weighting) were made with all folds in view and are therefore not blind; the $\kappa(T_e)$ assessment rests on fixed- β grid profiles with the remaining parameters refit at every grid point (within-run identifiability) and on two-run LORO transfer fits. A final fit on all runs provides the deployed model.

5 Results

5.1 Prediction accuracy

Table 1 and Figs. 2–3 give the headline result, and the added references sharpen what it means. The trivial gap baseline $T_e = T_v - c_f$ is within 3 K everywhere – the signal rises some 60 K, so all sub-3 K numbers should be read against that anchor – and it beats every nonlinear black box on the upward-extrapolation fold. What enables extrapolation is *structure*, not physics specifically: every linear two-state form fitted by output-error shooting extrapolates – the physical ODE at 0.41/0.26/0.73 K, the one-node ablation at 0.39/0.24/0.71 K, and an unconstrained per-fluid linear state-space with no physical parameterisation at all at 0.28/0.30/0.68 K – while free-form learners (MLP, GP, degree-2 SINDy) fail at 23 K to 87 K. The ridge-fitted linear state-space sits between (2.7 K to 5.8 K) because its rates come from noisy finite differences (errors in variables), not from shooting. The physical parameterisation is therefore unnecessary for *accuracy*: the free linear form matches or beats it in every fold; it is retained for parsimony (9 shared parameters against 3×8 free ones, fitted per fluid without cross-fluid borrowing) and for the interpretable R^* it exposes. The GRU, exploiting temporal structure, is the best black box on extrapolation (5 K) but still $\sim 6 \times$ worse than the structured fits. The ODE shot is slightly better than the PINN in every fold (0.41/0.26/0.73 vs. 0.45/0.35/0.85); for this fully-observed 9-parameter system the PINN machinery adds nothing measurable over classical shooting – the honest conclusion, not a surprise. Seed dispersion of the PINN is ≤ 0.03 K across ten seeds per fold.

Condenser T_c RMSE is 0.2 K to 1.5 K for all physics fits (T_c moves only 2 K to 5 K).

5.2 Structure ablation: is the second node needed?

The one-node model (5 parameters, no condenser node) matches the two-node fits on T_e everywhere (0.39/0.24/0.71 K vs. 0.41/0.26/0.73 for shooting). For evaporator prediction alone, the second node, r , d and T_{cool} earn nothing – the two-node structure is retained because it additionally predicts the condenser channel and is validated by the adiabatic consistency check below, and

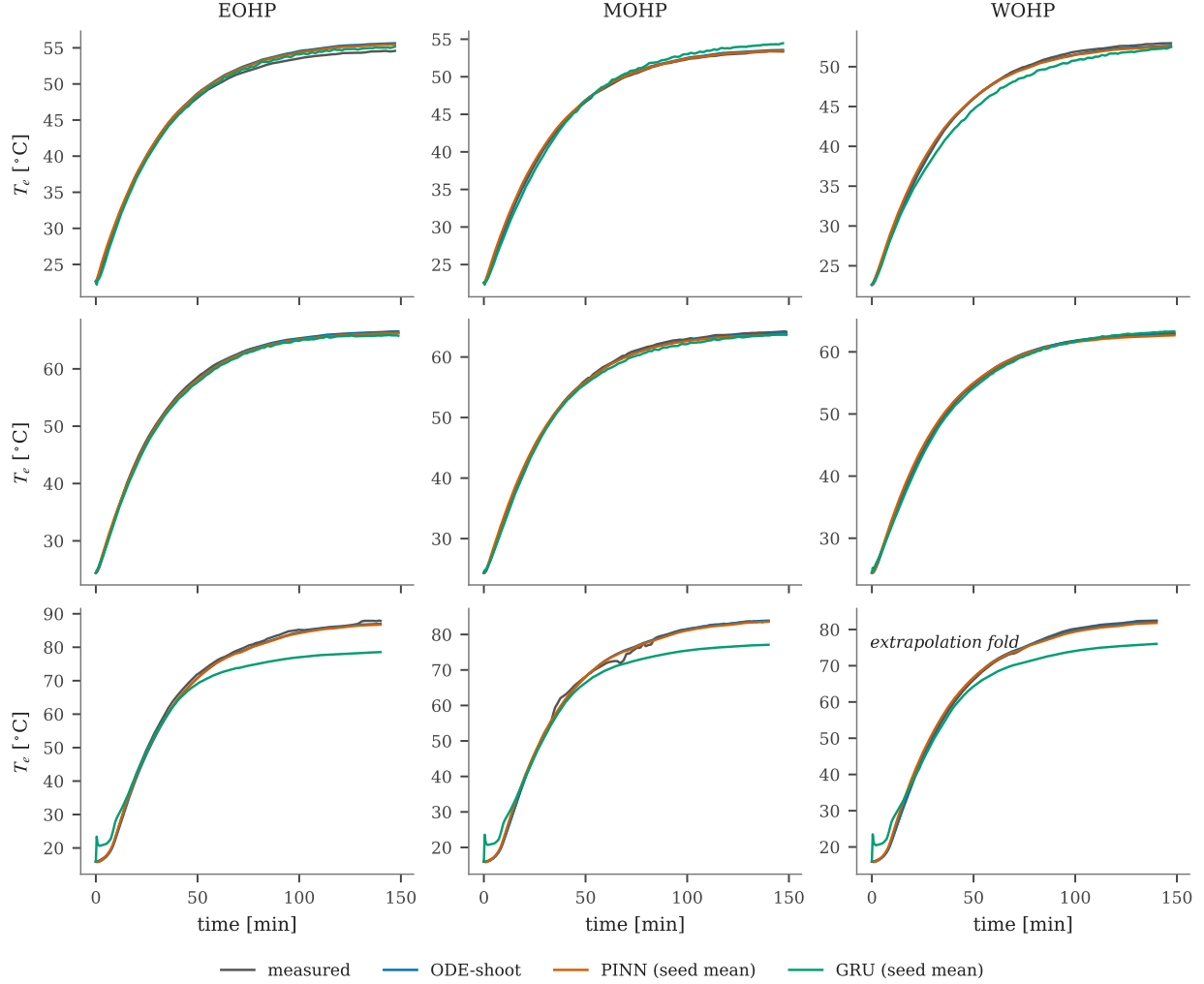


Figure 2: LORO held-out evaporator predictions (all folds and fluids): measured (black), ODE shooting (blue), PINN-const seed-mean (vermillion), GRU seed-mean (green). Run 3 is the extrapolation fold; black boxes degrade there, structured fits do not.

because its parameters carry the physical interpretation; but no accuracy claim rests on the extra node.

5.3 Identified parameters and R^*

R^* (Table 2, Fig. 4) ranks EOHP > MOHP > WOHP in every fit and every fold. It is a ratio of fitted rates and we do not attribute it to a specific resistance: the decomposition between vessel–evaporator contact resistance and pipe resistance is not identifiable here, and the source study reports fluid-dependent contact resistance. Quantitatively, in the final shooting fit $\ln R_E^*/R_W^* = 0.83$ splits as $\ln(a_E/a_W) = +0.54$ minus $\ln(\kappa_E/\kappa_W) = -0.29$: about two-thirds of the gap sits in the inlet coupling a_f and one-third in the pipe conductance κ_f (PINN: 60/40; runs/outer/rstar_decomposition.log). These shares are per-fluid ratios of fitted rates, hence invariant to the joint-rescaling sloppiness of Fig. 5, but they remain a decomposition of fitted rates, not measurements of separate physical resistances. In steady state $R^* = (T_e - T_c)/(T_v - T_e)$, and plateau ratios computed directly from the

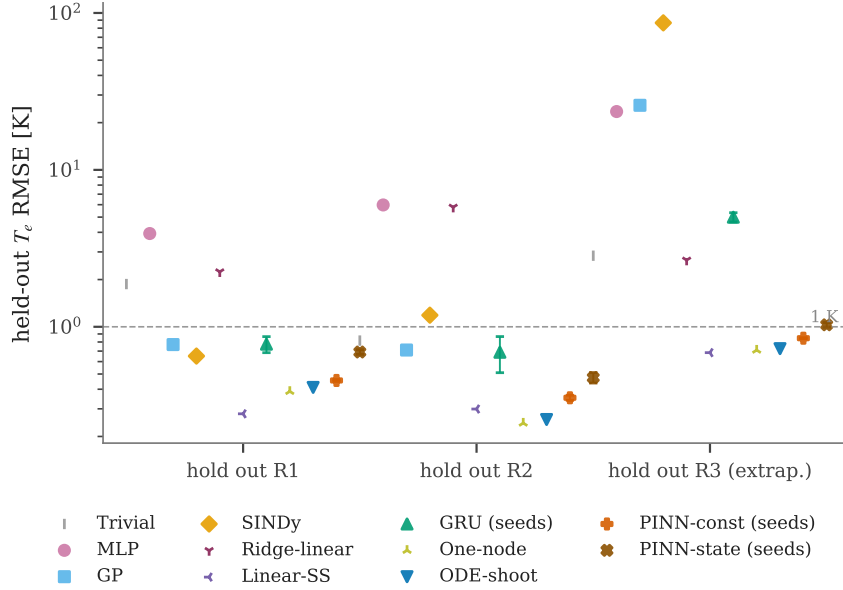


Figure 3: Held-out T_e RMSE by method and fold (mean over fluids; seed sd where applicable; log scale).

data give 9.9/12.4/14.6 (EOHP), 7.1/7.6/7.2 (MOHP), 6.4/6.1/6.1 (WOHP) for Runs 1–3 – the dynamics-based fit reproduces the plateau ratios and smooths their run-to-run drift. That drift is informative: EOHP’s ratio rises $\sim 48\%$ with heating level while methanol and water stay flat, consistent with the early, sharp ethanol dry-out reported in the source study (onset 70/83/98 W for ethanol vs. 105/135/160 W gradual onset for methanol and water); the constant- κ model absorbs this as a cross-run power-level effect ($\pm 20\%$ on κ) and cannot represent dry-out onset. R^* is invariant to the sampling interval ($DT_S = 1$ vs 5 reruns give identical rankings and near-identical values), and robust to the choice of vessel thermocouple driver (outer vs. mean-of-four changes absolute values by up to $\sim 12\%$ but never the ranking or the accuracy). The absolute rate parameters (a_f, κ_f) are constrained only to within a factor ~ 2 at the sub-kelvin level – rescaling both jointly leaves RMSE at 0.40/0.65 K for $1 \times / 2 \times$ (Fig. 5) – while their ratio is far stiffer across methods, seeds and folds; this is why R^* , not the rates, is the reportable quantity.

Jackknife-over-series refits of the shooting model (9 leave-one-series-out fits; runs/outer/jackknife.csv) give series-level uncertainty on the ordering: $\log R_E^* - \log R_M^* = 0.62 \pm 0.11$ and $\log R_M^* - \log R_W^* = 0.21 \pm 0.03$ (5.4 and 8.2 σ ; 95% t -intervals with 8 degrees of freedom [0.36, 0.89] and [0.15, 0.27], both excluding zero), i.e. the fluid ordering survives resampling at every level we can test, though with one shared driver per run these are series-level, not experiment-level, intervals.

5.4 Negative and corrected results

$\kappa(T_e)$: identifiable within runs, not transferable across them. An earlier draft of this analysis reported a per-fluid Arrhenius-type $\kappa_f(T_e) = k_{0f}e^{\beta_f(T_e-50)}$ as “not identifiable”; that verdict was an optimiser artefact (the within-run fits initialised β at exactly zero with a relative finite-difference step, which stalls there). Fixed- β grid profiles with the remaining five parameters refit at every point (runs/outer/kappa_te_profile.log) show β is clearly identifiable *within* a run: ethanol improves within-run RMSE by 67–71% at $\hat{\beta} \approx +0.03$ (19% in Run 3), water by 45–48% at $\hat{\beta} \leq -0.02$ (grid edge), methanol not at all ($\hat{\beta} \approx 0$) – a sign pattern consistent with the cross-run κ drift. What

Table 1: Held-out evaporator T_e RMSE (mean over fluids, K) against raw data; mean $\pm$ sd over seeds. Run 2 is interpolation; Runs 1 and 3 are downward/upward extrapolation in heating level.

	Run 1 (down extrap.)	Run 2 (interp.)	Run 3 (up extrap.)
Trivial ($T_v - c_f$)	1.87	0.82	2.84
MLP	3.93	5.98	23.54
GP (narrow / wide)	0.77 / 0.58	0.71 / 0.74	25.80 / 25.81
SINDy (deg 2)	0.65	1.19	86.49
Ridge-linear	2.24	5.78	2.66
Linear-SS (3×8 params)	0.28	0.30	0.68
GRU (3 seeds)	0.77 ± 0.09	0.69 ± 0.18	4.99 ± 0.34
One-node (5 params)	0.39	0.24	0.71
ODE-shoot (9 params)	0.41	0.26	0.73
PINN-const (10 seeds)	0.45 ± 0.01	0.35 ± 0.01	0.85 ± 0.01
PINN-state (10 seeds)	0.69 ± 0.02	0.47 ± 0.04	1.03 ± 0.03

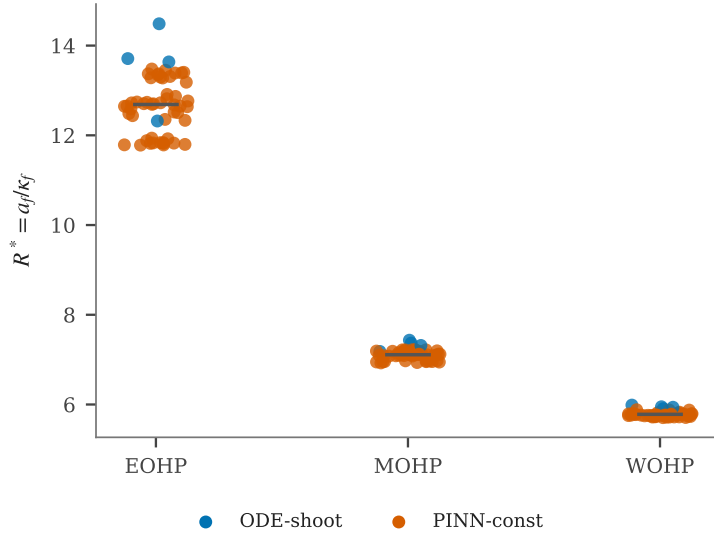


Figure 4: Dimensionless coupling ratio $R^* = a_f / \kappa_f$ from every fit (all LORO folds and seeds).

fails is *transfer*: fitted on two runs with all parameters shared and applied to the held-out run, the training $\hat{\beta}$ collapses toward zero and leaves held-out RMSE unchanged or worse in eight of nine folds (WOHP Run 1: 0.15 K at $\beta = 0$ vs. 0.52 K at the training $\hat{\beta} = -0.005$; one fold improves, EOHP Run 3: 0.77 $\rightarrow$ 0.50 K), and the joint PINN-kTe fit likewise degrades the extrapolation fold (0.85 $\rightarrow$ 1.19 K). The temperature dependence is thus run-specific – a heating-level effect, not a transferable fluid property – so the deployed model keeps κ constant; extrapolating a β identified at one power level to another remains open. **Per-fluid d unnecessary.** Freeing the condenser loss rate per fluid yields ≤ 0.02 K gains on interpolated folds and none on extrapolation; learned d ratios span only ~ 1.2 .

5.5 Adiabatic-channel consistency check

The adiabatic thermocouples were never used in fitting. A static mixture of the two node temperatures, $T_a \approx w_1 T_e + w_2 T_c$ (no intercept), reproduces them in-sample at 0.1 K to 0.4 K (vs. 19 K

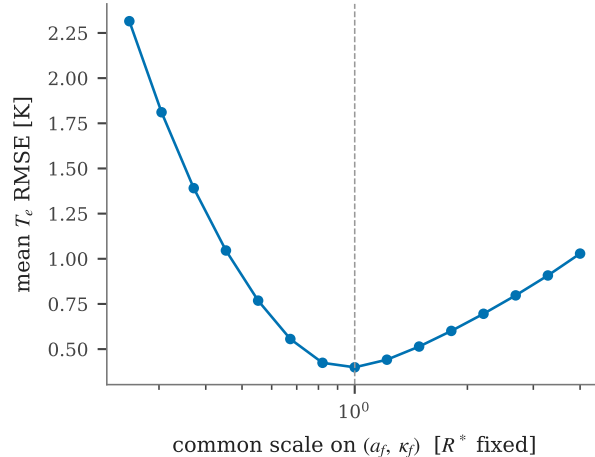


Figure 5: Identifiability: rescaling (a_f, κ_f) jointly (fixed R^*) barely moves the held-out RMSE – the rates are sloppy, their ratio is not. The absolute rates still set scenario timescales; that band is propagated in the scenario metrics below.

Table 2: Dimensionless coupling ratio $R^* = a_f/\kappa_f$ (mean $\pm$ sd over all LORO + final fits). The ODE-shoot jackknife-over-series row gives 68% intervals: $R^* = 13.6 \times / \div 1.12$ (EOHP), $7.3 \times / \div 1.03$ (MOHP), $5.9 \times / \div 1.01$ (WOHP).

	EOHP	MOHP	WOHP
ODE-shoot	13.54 ± 0.90	7.33 ± 0.11	5.95 ± 0.03
PINN-const	12.63 ± 0.56	7.09 ± 0.09	5.76 ± 0.03

to 47 K for T_v alone), with $w_1 + w_2 \approx 0.96$ – 0.99 and w_1 fluid-ordered (E $0.14 < M$ $0.24 < W$ 0.34 : the adiabatic wall sits nearer the condenser for water). Under LORO, held-out T_a is predicted at 0.27 K to 0.45 K for EOHP/WOHP and 0.6 K to 3.5 K for MOHP (Run 3 is the hard case); chaining model-predicted T_e, T_c (Fig. 6) stays at 0.14 K to 0.76 K for E/W and 4.25 K for MOHP Run 3. We present this as a consistency check, not a structural test: two free weights can interpolate any profile lying between the two nodes, and the chained version scoring slightly better than the measured-input version (0.14 vs. 0.27 K for EOHP Run 1) shows the metric is insensitive at this level. A one-parameter variant ($w_2 = 1 - w_1$, $w_1 = 0.12/0.20/0.32$ for E/M/W) performs the same or better on EOHP/WOHP (0.17 K to 0.58 K held-out), confirming the check constrains little beyond “ T_a lies between the nodes”. It does establish that the fitted two-node states are mutually consistent with an untrained measurement.

5.6 Exported model and a model-implied scenario illustration

The fitted right-hand side (not the PINN network) is exported as ONNX and streams held-out Run 3 at 0.85/0.79/0.88 K (E/M/W) in a replay loop (Fig. 7). As an *illustration of what the identified model implies* – not a validated design tool – we drive it with three waveform families (ramp-and-hold to 60/75/90 °C at three rates; 4 on/off thermostat cycles; a 10 K 60 s disturbance slug), all inside the fitted temperature envelope. Within this linear model the fluids differ mainly as a first-order lag: on tracking metrics (deviation from setpoint, vessel–evaporator lag, settling) EOHP wins in 100% of (fit, scenario) cases (mean lag 4.0 vs. 8.1 K for WOHP); on ripple rejection

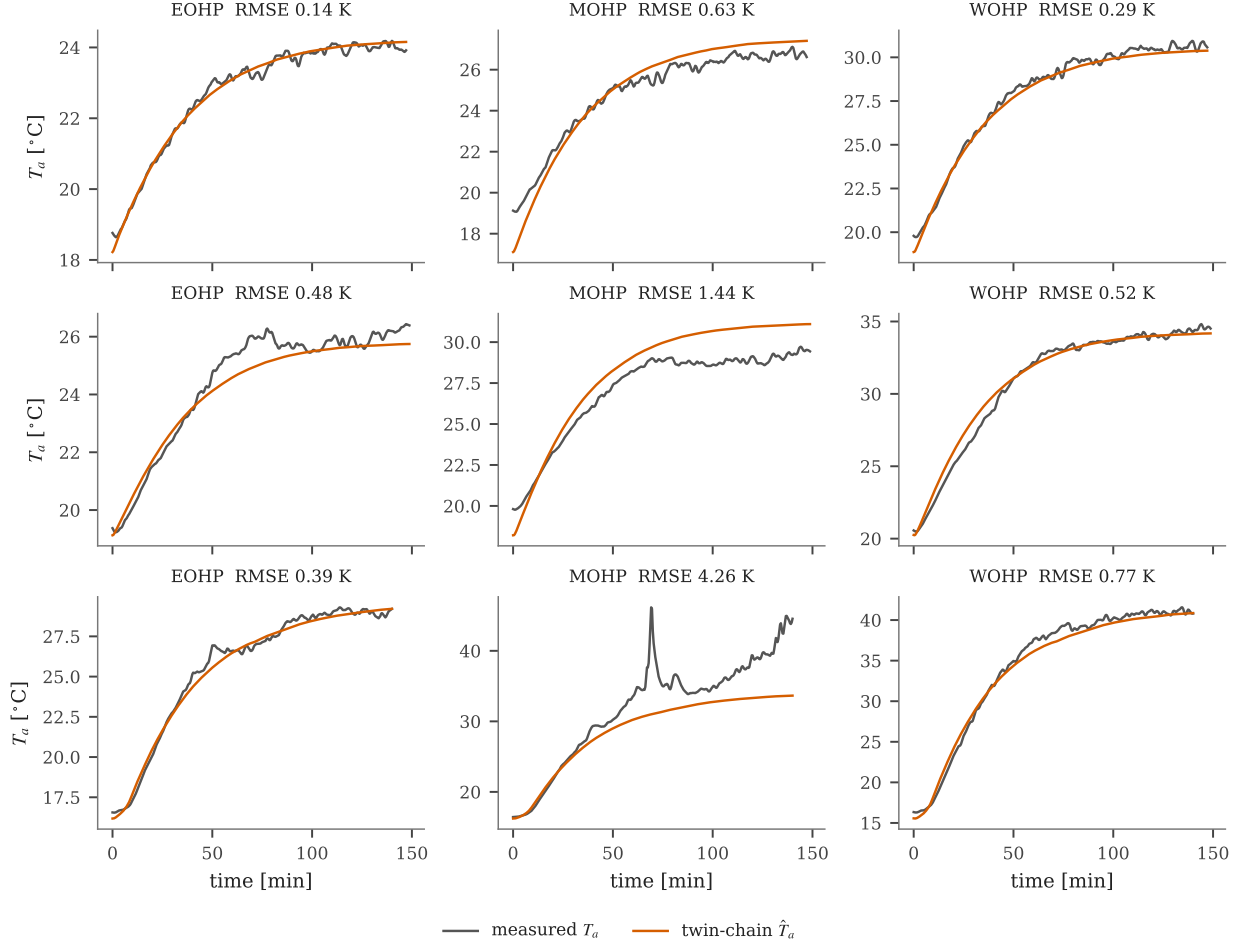


Figure 6: Held-out adiabatic channel (never fitted): measured T_a vs. the chained prediction $T_a = w_1 T_e^{\text{pred}} + w_2 T_c^{\text{pred}}$, with (w_1, w_2) fitted on the training runs only. EOHP and WOHP agree at sub-0.5 K RMSE; MOHP Run 3 is the identified hard case.

WOHP wins at the identified rates (cyclic-duty ripple 28–49 vs. 31 K to 55 K across setpoints; slug response 1.8 vs. 2.8 K at 60 °C), the same bandwidth trade-off read from the other end. The ordering is monotone in R^* , though with three fluids two of six orderings match by chance, so this is consistent-with, not evidence for, R^* as a selection dial. Two uncertainty checks qualify the case study (runs/outer/scenario.bands.log). Across the 10 seeds of the final fit the scenario metrics vary by only ± 0.03 K to 0.15 K, far below the inter-fluid gaps. But the absolute rates are themselves sloppy (Fig. 5): replaying all scenarios with (a_f, κ_f) jointly rescaled $\times / \div 2$ – which barely moves held-out RMSE – and through the 9 leave-one-series-out jackknife refits leaves every tracking metric and its ordering unchanged (EOHP best in 100% of variants; the quasi-static lag $T_v - T_e = R^*(T_e - T_c)$ is rate-invariant by construction), while the genuinely dynamic ripple metrics move by up to ~ 6 K, and inside that band the EOHP–WOHP ripple ordering is not separable (best-fluid share 60/40 across variants). The tracking advantage is robust to everything we can vary; the ripple advantage is real at the identified rates but not credibly resolvable given how loosely those rates are identified. Because the model contains no heat-flux limit or dry-out onset – the phenomenon the source study identifies as decisive for fluid choice, and which hits ethanol

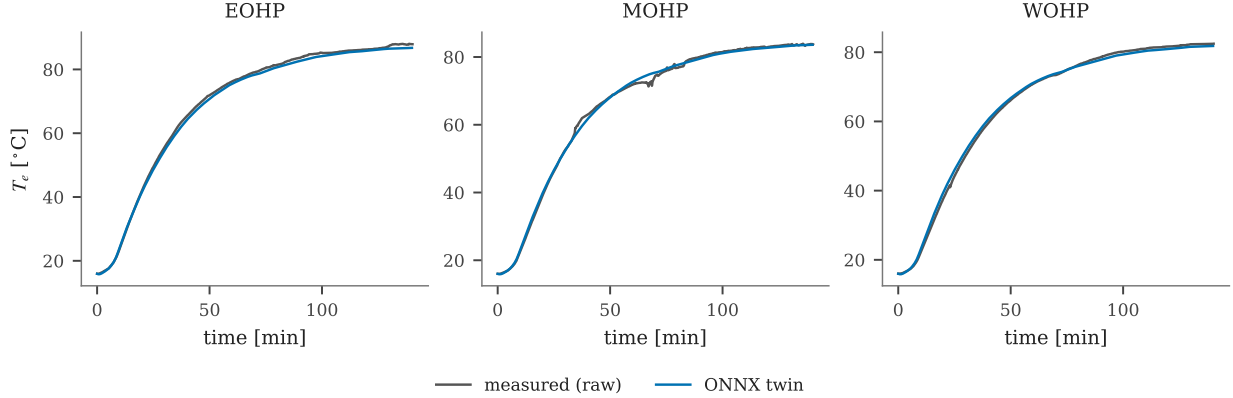


Figure 7: Streaming ONNX replay of held-out Run 3 (fold-3 model, never trained on Run 3).

first – these illustrations must not be read as fluid recommendations; they show what the identified linear dynamics imply, no more.

5.7 What the seed ensemble does and does not measure

The 10-seed ensemble spread of held-out rollouts is 0.01 K to 0.05 K (2 sd), while actual held-out residuals are 0.25 K to 0.89 K (Fig. 9): nominal 95% bands cover 0–34% of the raw data. Seed dispersion quantifies parameter (identifiability) uncertainty only; the residual is dominated by structural and measurement error. We therefore report seed bands as parameter uncertainty, not as full predictive bands (conformal augmentation is future work).

6 Discussion

The consistent picture across Table 1 is that extrapolation is enabled by imposing a correct (low-dimensional, linear) structure and fitting it by output-error shooting, not by physics-inspired optimisation machinery and not by the physical parameterisation itself: the shooting fit of the same equations matches or beats the PINN, a one-node variant does as well on the evaporator, and an unconstrained per-fluid linear state-space with no physical content does slightly better still (0.28/0.30/0.68 K). The physical model’s yield is accordingly not accuracy but parsimony (9 shared parameters against 3×8 fitted independently per fluid) and interpretability – the dimensionless R^* , which the free form does not expose. This is a caution for the PINN literature on small lumped systems – where the classical shooting fit is trivially available – and a clarification of what grey-box structure buys in thermal identification. It is also a caution about black-box surrogates for design: every generic learner, including a recurrent model trained on dynamics, failed the extrapolation question that the structured model answers at sub-kelvin accuracy – though a trivial constant-gap predictor already achieves 3 K, so the practical margin over naive baselines is modest and should be stated. The scientific yield is a dimensionless coupling ratio per fluid, robust to sampling, seed and method, that orders the fluids and tracks the ethanol dry-out signature reported in the source data; the adiabatic channel offers a weak but free consistency check on the identified states.

Limitations. One geometry; three fluids; effectively one independent driver per fold (the three fluids share the vessel trace); structure-selection choices were made with all folds in view; slow heating ramps (quasi-steady oscillation statistics are not resolved at 5 s); equal vessel–evaporator coupling assumed across pipes, though the source reports fluid-dependent contact resistance, so the

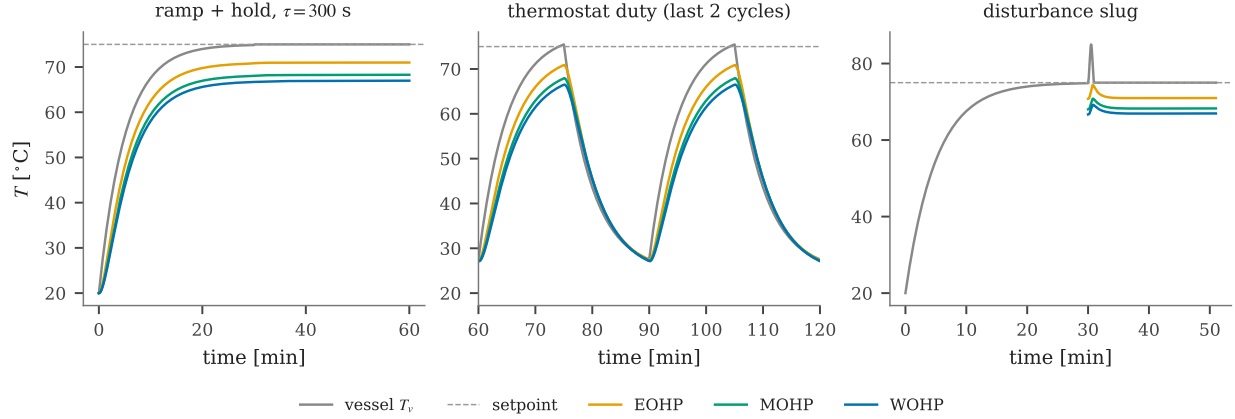


Figure 8: Model-implied scenarios at a 75 °C setpoint (ONNX rollouts, final fit). (a) Ramp-and-hold ($\tau = 300$ s): EOHP tracks the vessel tightly (lowest lag, fastest settle); WOHP lags most. (b) Thermostat on/off duty (last two cycles): WOHP shows the smallest evaporator ripple. (c) 10 K, 60 s disturbance slug: WOHP transmits the least disturbance. EOHP’s tracking advantage is robust to rate rescaling and jackknife refits; the ripple ordering is not separable within the rate-identifiability band (text).

R^* ordering cannot be attributed to the pipe alone; the constant-parameter model cannot represent dry-out onset, which the source study identifies as decisive for fluid choice; absolute fluxes are not identifiable from these records (the dataset’s derived flux constant $h = 2100 \text{ W m}^{-2} \text{ K}^{-1}$ could not be traced to a source, and the derived vessel-flux columns reproduce the cylindrical-wall conductance but are thermocouple-resolution-limited); the known MOHP Run 3 flow-state change and hot-slug transient are left in (excluding ± 150 s windows changes that fold’s RMSE only marginally). Results do not extend to hotspot-flux, dry-out-limited or geometry-optimisation questions.

7 Conclusion

A two-node, vessel-driven grey-box model of a helically coiled OHP, identified by PINN or plain ODE shooting from a public three-fluid dataset, predicts unseen runs at 0.24 K to 0.85 K, including extrapolation ± 23 K in heating level where all nonlinear black-box baselines fail; the identified right-hand side exports as ONNX, remains consistent with an untrained adiabatic channel, and yields a dimensionless coupling ratio R^* that is robust across methods, seeds and sampling, and whose ethanol drift matches the dry-out signature reported in the source data. The honest accounting: structure, not the optimiser, carries the accuracy; the second node is justified by the auxiliary channels rather than evaporator accuracy; and the model’s scenario implications are illustrations within its validated envelope, not fluid recommendations.

Data availability

All results derive from the public dataset of [1] (Mendeley Data, DOI 10.17632/wnf5jwzp3c.3, CC BY 4.0). Code, fitted parameters, ONNX models and result files are archived at [Zenodo DOI upon publication].

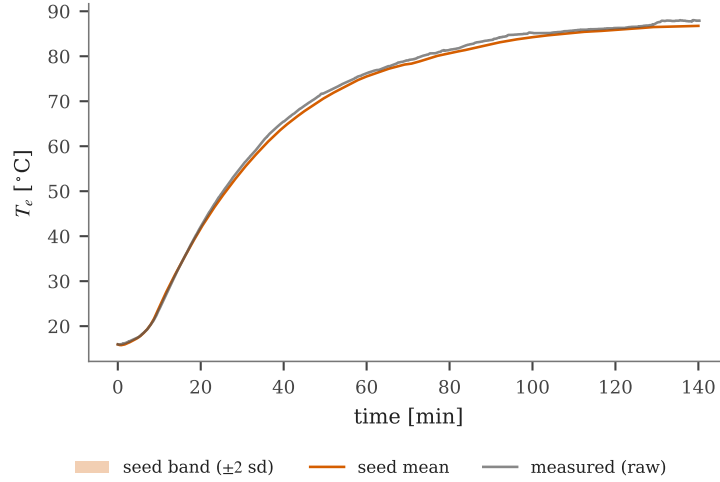


Figure 9: Seed-ensemble band (10 PINN seeds, mean $\pm$ 2 sd) on the held-out Run 3 EOHP evaporator temperature: the parameter band is far narrower than the residual – seed spread quantifies identifiability, not total error.

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